AXLERO SOLUTIONS • PROJECT 2

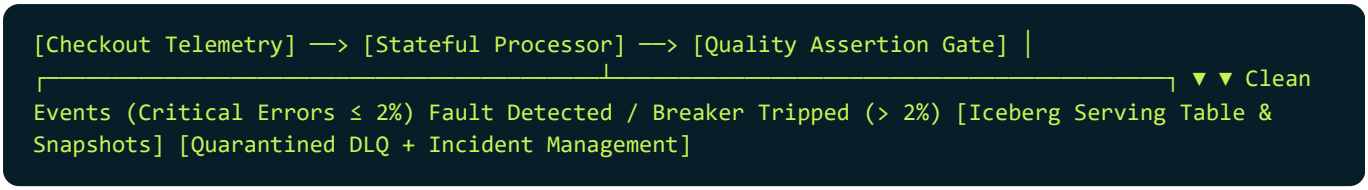
IceStream Observability

Viva & Reviewer Guide

1. Executive Summary (60-Second Pitch)

Elevator Pitch: IceStream is a real-time data-quality control plane for streaming checkout telemetry. Every transaction is governed and validated against strict quality contracts before entering the serving table. If the rolling critical error rate exceeds **2%**, an automated circuit breaker trips, isolates bad records to a Dead Letter Queue (DLQ), triggers live WebSocket alerts, and creates an auditable incident. Once clean batches arrive, it automatically recovers. Point-in-time immutable snapshots allow operators to perform time-travel audits.

2. Pipeline Architecture & Data Flow



Step-by-Step Flow:

1 Ingress & Stateful Stream Validation

Incoming order events arrive with subtotal, tax, region, currency, and schema version. The engine evaluates each record against 6 governed assertions in memory.

2 Rolling 2% Circuit Breaker Protection

The control plane monitors a sliding 100-event window. If critical quality failures cross 2% (`error_rate > 0.02`), the circuit trips from *closed* to *open*.

3 Dual-Route Isolation (Serving vs. DLQ)

Clean records commit to the lakehouse storage sink. Quarantined records route to SQLite Dead-Letter Storage with exact failure reasons—bad data never infects downstream analytics.

4 Automatic 5-Batch Hysteresis Recovery

To avoid flapping, recovery requires 5 consecutive clean probe batches. Once verified, the circuit closes automatically, or an operator can trigger a manual reset.

5 Lakehouse Snapshots & Time-Travel

Every 24 accepted events commit an immutable snapshot watermark. Operators can inspect point-in-time historical tables to audit historical state.

3. Governed Data Quality Rules Catalogue

Rule ID	Rule Name	Field	Severity	Quality Contract / Assertion Logic
DQ-001	Tax is present	`tax_amount`	`CRITICAL`	Checkout events must contain a valid, non-null tax amount.
DQ-002	Tax is plausible	`tax_amount`	`CRITICAL`	Tax must be non-negative and ≤ 35% of subtotal.
DQ-003	Schema is governed	`schema_version`	`CRITICAL`	Only approved `checkout.v1` schema may enter serving.
DQ-004	Currency is supported	`currency`	`WARNING`	Currency must match supported settlement codes (USD, EUR, INR, GBP).
DQ-005	Subtotal bounded	`subtotal`	`CRITICAL`	Subtotal must remain inside operational threshold (≤ $50,000).
DQ-006	Region is known	`region`	`WARNING`	Region must map to recognized geographic sales hierarchy.

4. Live Demo Walkthrough (For the Reviewer)

1 Login & Dashboard Overview

Open `http://localhost:3000`. Log in with Username: **Abhishek001**, Password: **IceStream@2026**. Point out the 4 live metrics (Events, Quality Score, Quarantined, Throughput) and the React Flow lineage canvas showing a healthy green path.

2 Controlled Fault Injection

Click **"Null tax burst"** or **"Schema drift"** in the right-side Controlled Test panel. Show the reviewer how the Quality Score drops, the circuit breaker opens, and the Serving node turns red while Quarantine becomes active.

3 Audit the Incident Log

Navigate to the **Incidents** tab. Show the newly opened incident badge, violation details (e.g., `DQ-001`), and timestamp. Explain that rejected payloads are preserved in DLQ for forensic debugging.

4 Circuit Recovery Demonstration

Click **"Manual Reset"** or allow 5 consecutive clean batches to auto-recover the circuit. The lineage edges turn green again.

5 Lakehouse Snapshots & Time Travel

Switch to the **Snapshots** tab. Click on a snapshot ID to inspect the immutable JSON watermark and record payload at that exact historical boundary.

5. High-Scoring Viva Q&A Cheat Sheet

Q1: Why pre-ingestion quality gates instead of post-write batch validation?

Post-write validation lets corrupted records pollute analytics dashboards and downstream machine learning models before alerts trigger. IceStream acts as an in-line gate, stopping unsafe writes before commitment.

Q2: Why use a rolling window instead of a cumulative lifetime average?

A lifetime average dilutes recent degradation (e.g., 50 failures out of 100,000 events is only 0.05%). A rolling 100-event window detects sudden spikes instantly and clears once the stream stabilizes.

Q3: What is the difference between Quarantine (DLQ) and deleting records?

Deleting bad data loses forensic auditability and financial reconciliation evidence. A Dead Letter Queue preserves the exact malformed payload and rule failure metadata for replay.

Q4: How does this prototype map to production cloud infrastructure?

In production, the generator maps to **Apache Kafka / Redpanda**, the stateful processor maps to **Apache Flink** (with side-outputs for DLQ), and the local SQLite storage maps to **Apache Iceberg tables on AWS S3 / MinIO**.